Ultrafast X-Ray Experiments

Virtual Experiments at the X-Ray Laser European XFEL
Versuch EXP21

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Datenstruktur (beam imaging and xray eye): 420x420 pixel, bei biu screen dim von 10mm x10mm, bei xray eye screen 2mm x 2mm

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monochromator focusing optics: mirrors, power slits, solid attenuators, refractive Lenses, X-ray eye

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diffraction gratings

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Quelle Versuchsmappe:[1]

4. Experimental Procedure

4.1. Preparation of the Experiment

4.1.1. Alignment and Characterization of the X-ray Beam

Before doing anything on a beam line it's important to characterize it.
As the absorption edge for the Fe K$\beta_{1,3}$ line of 7065 eV, the energy of the xray beam is set to 7400 eV, to avoid the ripples near the reaction (siehe **??**) but without too strong of a loss in intensity. To measure

the initial X-ray beam the power slits are fully opened and the following attenuator foils are used to bring the beam to an intensity just below saturatuion of the beam imaging unit 2 (BIU2):

The size and position of the x-ray is now measured using BIU2. Now the four-bounce crystal-monochromator is turned on and then off again to record the reduction of the spectral bandwidth. Then the dispersive single-shot spectrum analyser is used to record the spectrum of the pink X-ray beam with a bragg angle of 24°.

For the following experiment the monochromatic beam is used as it leads to a higher resolution than the pink x-ray, even though the intensity of the beam suffers.

The last step of diagnosing the beam line is using the refractive beryllium x-ray lenses to focus the beam at the imaging unit x-ray eye. It's important to appropiately adjust the attentuator foils as to not burn the imaging unit with a focussed beam.

With beryllium lenses 1, 2 and 5 (taken from **??**) and the attentuator foils listed here **??**, a beam with the FWHM of 13 um (siehe **??**) is achievd. The total focus length is calculated with $\delta \approx 7 \cdot 10^{-6}$ [**?**] and the properties of the lenses themself taken from **??**:

$$F = \frac{R}{2\delta N} \Leftrightarrow \frac{1}{F_{ges}} = \frac{2\delta N_1}{R_1} + \frac{2\delta N_2}{R_2} + \frac{2\delta N_5}{R_5}$$
$$\Leftrightarrow F_{ges} = \frac{1}{2\delta(\frac{N_1}{R_1} + \frac{N_2}{R_2} + \frac{N_5}{R_5})} \approx 4{,}76\,\mathrm{m}$$

—Notes—

X-ray energy $E_0 = 7400eV$ (has to be a good bit away from reaction edge of 7065 keV to be stable)

Alignment: attenuators to not burn through out screens (here: 4: B4C 0.8, 3: C 0.4, 2: Si 0.05, C 0.1) size and position of the beam plots (FWHM) (jitter neu!) pink vs mono beam 2D/3D plots

optical resolution: 25 um/px

Characterization: spectrum of the pink vs mono beam plots myb noch ex 7+8, aber idk

Focusing: plots of the focused beam size and position (FWHM)

4.1.2. Preparatory Estimates for the Experiment

Fun Facts: molar mass: $M([Fe(bipy)_3]^{2+}) \approx M([Fe(bipy)_3]) = 524,409\frac{g}{mol}$ concentration of the sample: $c = 0,380\frac{mol}{L}$ total attenuation of X-rays: attenuation comparison plot flow speed of the jet: $v_{jet} = 11\frac{m}{s}$

Laser: excitation wavelength: $\lambda_{pump} = 400nm$ (dazu alle fun facts mit MLCT Zuständen und so, und nicht overshooten lol) focussing, diameter of the focus Overfilling: beam sizes of the optical laser and the X-ray pulse relative to each other (plot beam diameter)

Spectromter: Van Hamos Spectrometer vs Johann Spectrometer (plot of the two geometries) crystal selection: $Si(531)$ vs $Ge(531)$ (plot of the two crystal reflections)

4.2. Performing the Experiment with Data Acquisition

5. Analysis and Interpretation

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7. Conclusion and Outlook

References

[1] C. Bressle, “Ultrafast x-ray experiments virtual experiments at the x-ray laser european xfel.”

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